

Skin Disease Detection - Final Model Comparison

- Dataset classes: chickenpox, eczema, ringworm
- Dataset split: train=379, validation=110, test=55 images
- Purpose: compare five coursework implementations on the same skin disease task.
- This project is experimental and is not a medical diagnostic tool.

Implemented Models

- Model A: Improved VGG-style PyTorch CNN with BatchNorm, Dropout, augmentation, AdamW, and class weights.
- Model B: Deeper plain PyTorch CNN with BatchNorm, global average/max pooling, AdamW, and no dropout.
- Model C: Simple two-convolution PyTorch CNN with Flatten and one hidden Linear layer.
- Model D: Fully connected PyTorch ANN/MLP over flattened image pixels.
- Model E: Simple scikit-learn raw-pixel LinearRegression baseline using one-hot targets and argmax.

Epochs and Training Setup

- Model A: configured for 60 epochs, best checkpoint at epoch 41.
- Model B: configured for 25 epochs, best checkpoint at epoch 22.
- Model C: ran for 30 epochs, best checkpoint at epoch 9.
- Model D: configured for 50 epochs, early-stopped after 11 epochs, best checkpoint at epoch 3.
- Model E: no neural training epochs; one LinearRegression fit on raw 8x8 pixel features.

Measured Results

- Rank 1 - Model B: test accuracy 74.55%, macro F1 0.7408, validation loss 0.7276.
- Rank 2 - Model A: test accuracy 69.09%, macro F1 0.6831, validation loss 0.8066.
- Rank 3 - Model C: test accuracy 56.36%, macro F1 0.5399, validation loss 1.3903, test loss 1.6501.
- Rank 4 - Model D: test accuracy 49.09%, macro F1 0.4399, validation loss 1.0031, test loss 1.0320.
- Rank 5 - Model E: test accuracy 45.45%, macro F1 0.4306, validation loss 1.1261, test loss 1.1966.

Conclusion

- Model B is the best implementation by accuracy, macro F1, and validation loss.
- Model A is the second-best model and remains a strong CNN baseline.
- Model C is simple and explainable but overfits strongly.
- Model D proves that an ANN can be trained, but flattened pixels generalize poorly.
- Model E is intentionally simple; linear regression is not a proper multiclass image classifier and should remain a weak baseline.